ML in Finance: Principal Component Analysis of Global Markets

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1 Introduction

Principal Component Analysis (PCA) is a foundational dimensionality reduction technique used in quantitative finance to identify latent drivers of asset returns. Whether analyzing the term structure of interest rates or the sector rotation of the S&P 500, PCA allows us to distill hundreds of correlated variables into a few interpretable factors.

2 Mathematical Foundation

2.1 Data Representation and Stationarity

To ensure the data is stationary, we transform the raw prices or yields:

- **Interest Rates:** We utilize *first differences* ($\Delta y_t = y_t - y_{t-1}$) because yields are already expressed in percentages.
- **Equities:** We utilize *percentage changes* or returns ($r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$) to normalize for absolute price levels.

2.2 The Covariance Matrix

PCA seeks to find the directions of maximum variance. Let X be the $T \times n$ matrix of centered changes. The sample covariance matrix Σ is:

$$\Sigma = \frac{1}{T-1} X^T X$$

2.3 Eigen-Decomposition

We diagonalize Σ to find the principal components:

$$\Sigma = Q \Lambda Q^T$$

where Λ contains the eigenvalues and $Q = [v_1, v_2, v_3]$ contains the eigenvectors.

3 Cross-Asset Interpretation

The first three principal components often explain over 95% of market movement.

Component	**Fixed Income Interpretation**	**Equity Interpretation**
PC1	Level (Parallel Shift)	Market Factor (Beta)
PC2	Slope (Twist)	Sector/Style Rotation
PC3	Curvature (Butterfly)	Specific/Thematic Clusters

Table 1: Comparison of latent factors.

4 Historical Case Studies

These interpretations were created by Google Gemini. As such, you need to analyze them carefully and assess their accuracy for yourself.

4.1 Evolution of the Interest Rate "Level" Factor (PC1)

A longitudinal analysis of the Scree plots in Appendix A reveals how the dominance of the Level factor (PC1) fluctuates with Fed policy and economic stability:

- **2000–2005 (81.2%):** High synchronization following the Dot-com bubble burst and subsequent easing cycle.
- **2005–2010 (60.9%):** A significant drop in PC1 dominance. As seen in Appendix A, the Slope factor (PC2) spiked to 26.3% variance as the curve twisted violently during the Global Financial Crisis.
- **2010–2015 (86.6%):** Maximum PC1 dominance. During the "Zero Bound" era, the entire curve was pinned by quantitative easing, leading to highly parallel movements.
- **2015–2020 (78.3%):** A return to moderate levels as the Fed attempted to normalize rates before the COVID-19 shock.
- **2020–2025 (73.2%):** While still dominant, high-dispersion events prevented the Level factor from reaching the levels seen in 2010.

4.2 The 2008 Global Financial Crisis (2005–2010)

During this era, the **Financials (XLF)** sector was the primary driver of market variance.

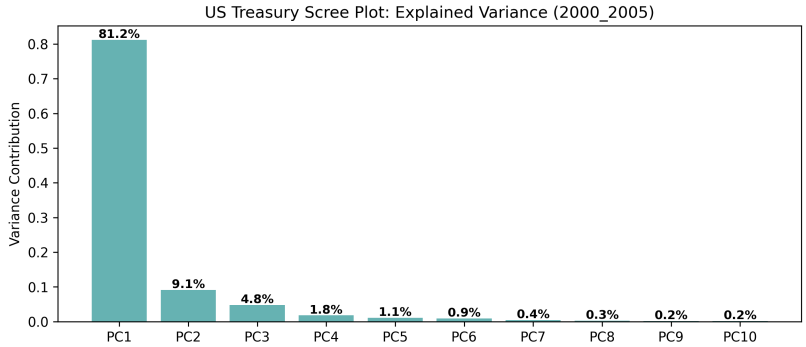
- **Market Sensitivity:** XLF exhibited the highest positive weight in PC1 (Market Beta), as seen in the Factor Loadings in Appendix D.
- **Sector Specifics:** The massive spike in XLF within PC2 (Value vs. Growth) highlights that the crisis was specifically a financial-sector event.

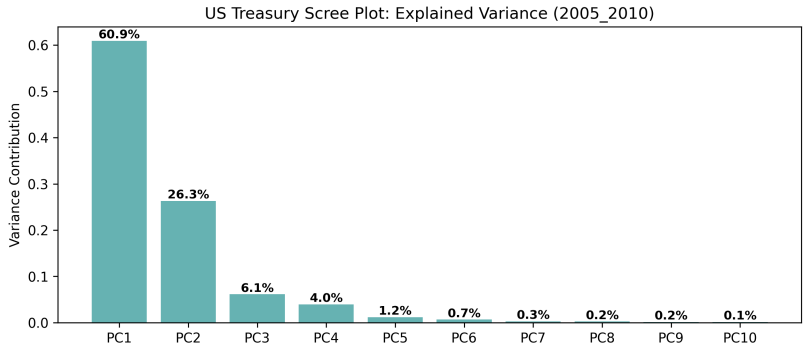
4.3 The Pandemic and Inflation Era (2020–2026)

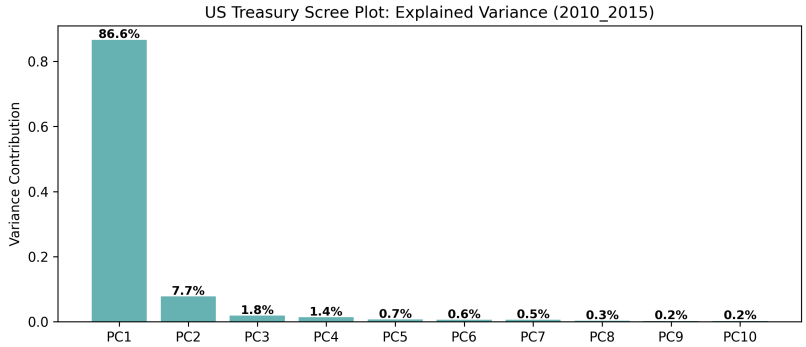
The current regime is defined by **Technology (XLK)** dominance and a clear Growth-led market structure.

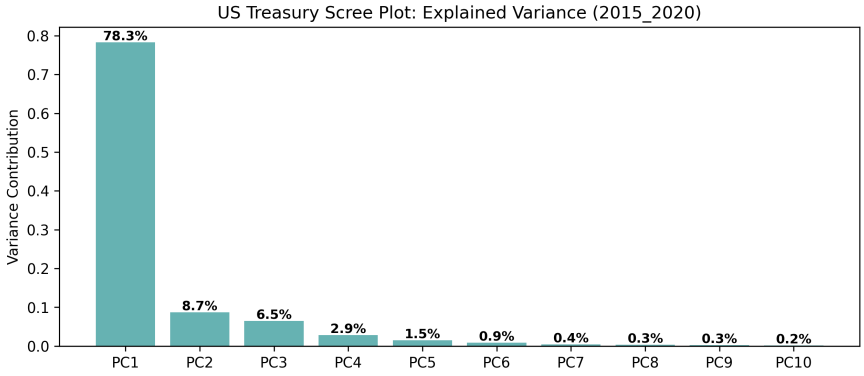
- **Tech as Beta:** In Appendix D, XLK shows extreme sensitivity in PC1, becoming a proxy for the overall market.
- **Growth vs. Value:** The extreme negative loading of XLK in PC2 illustrates its role as the engine for "Growth" trades.

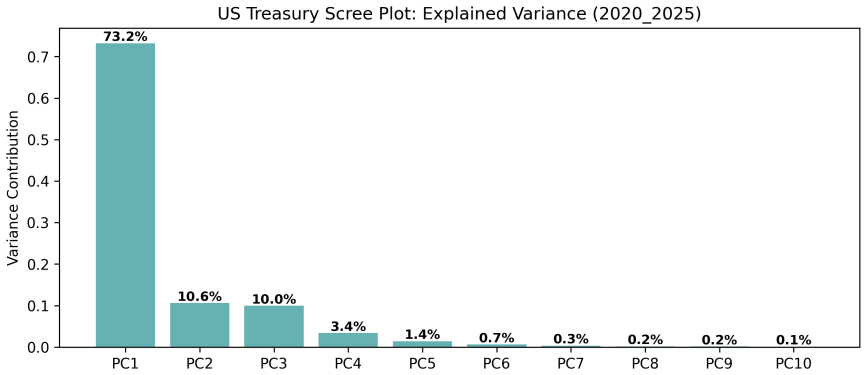
A Interest Rate: Scree Plots (2000–2025)



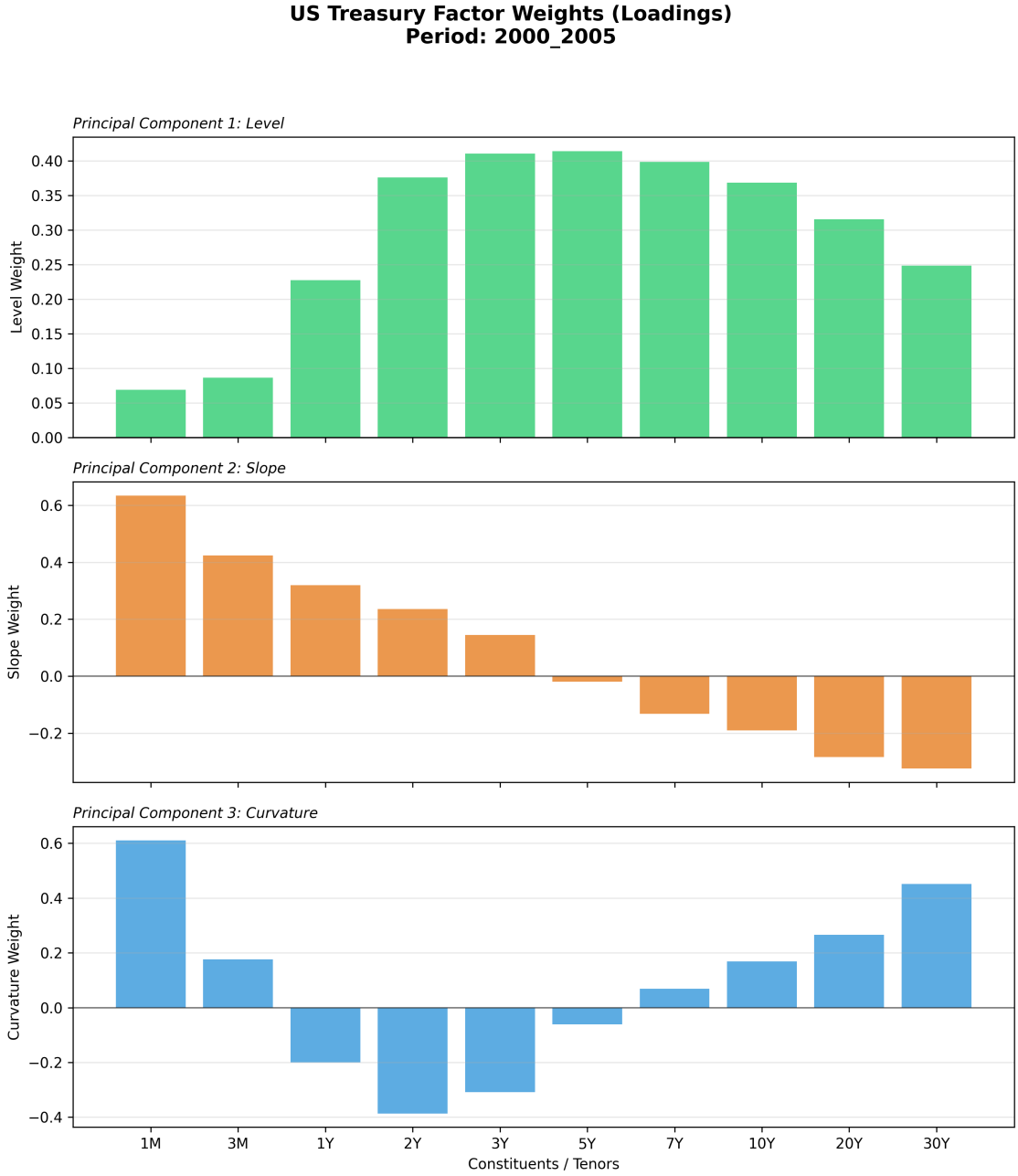


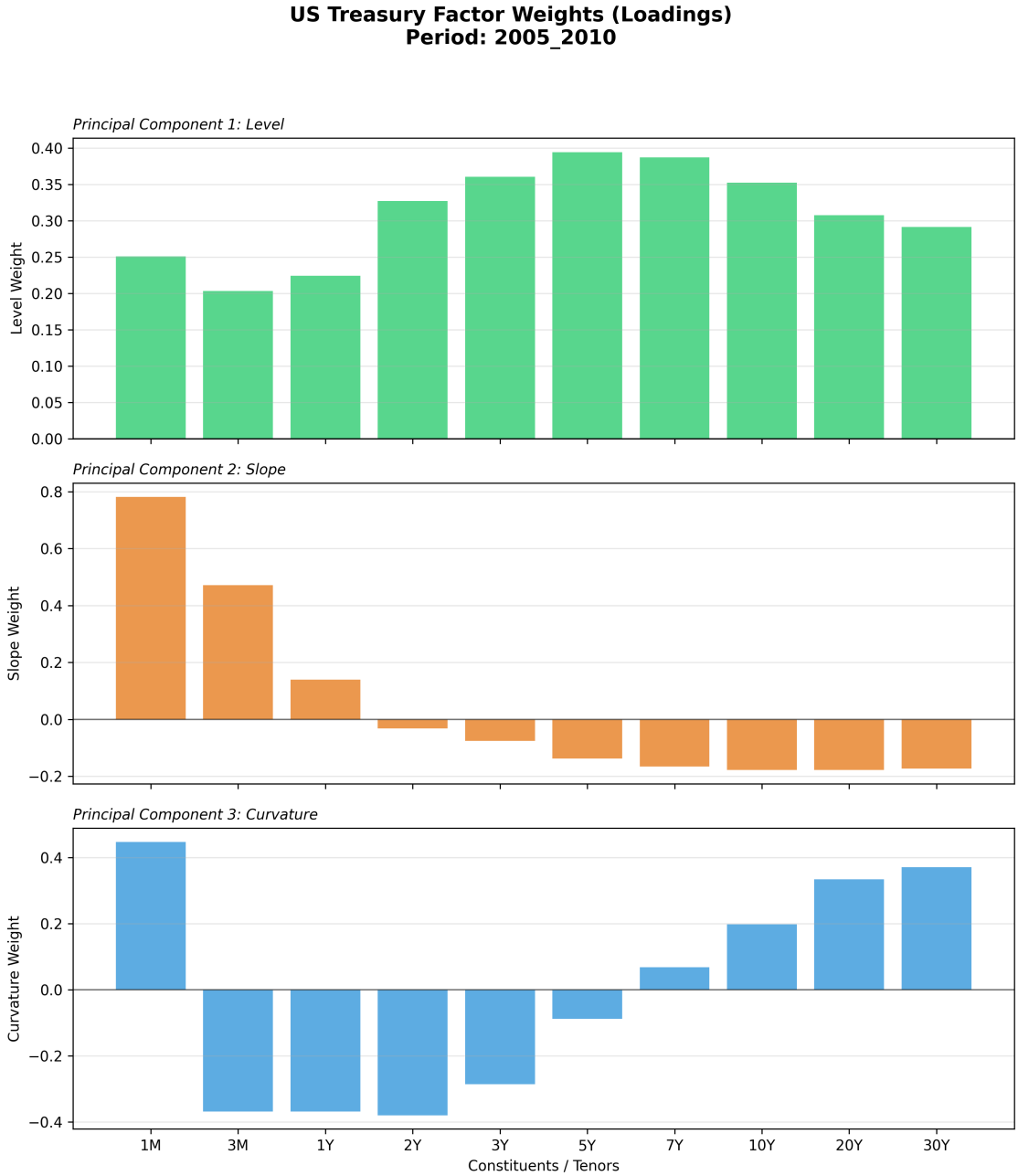


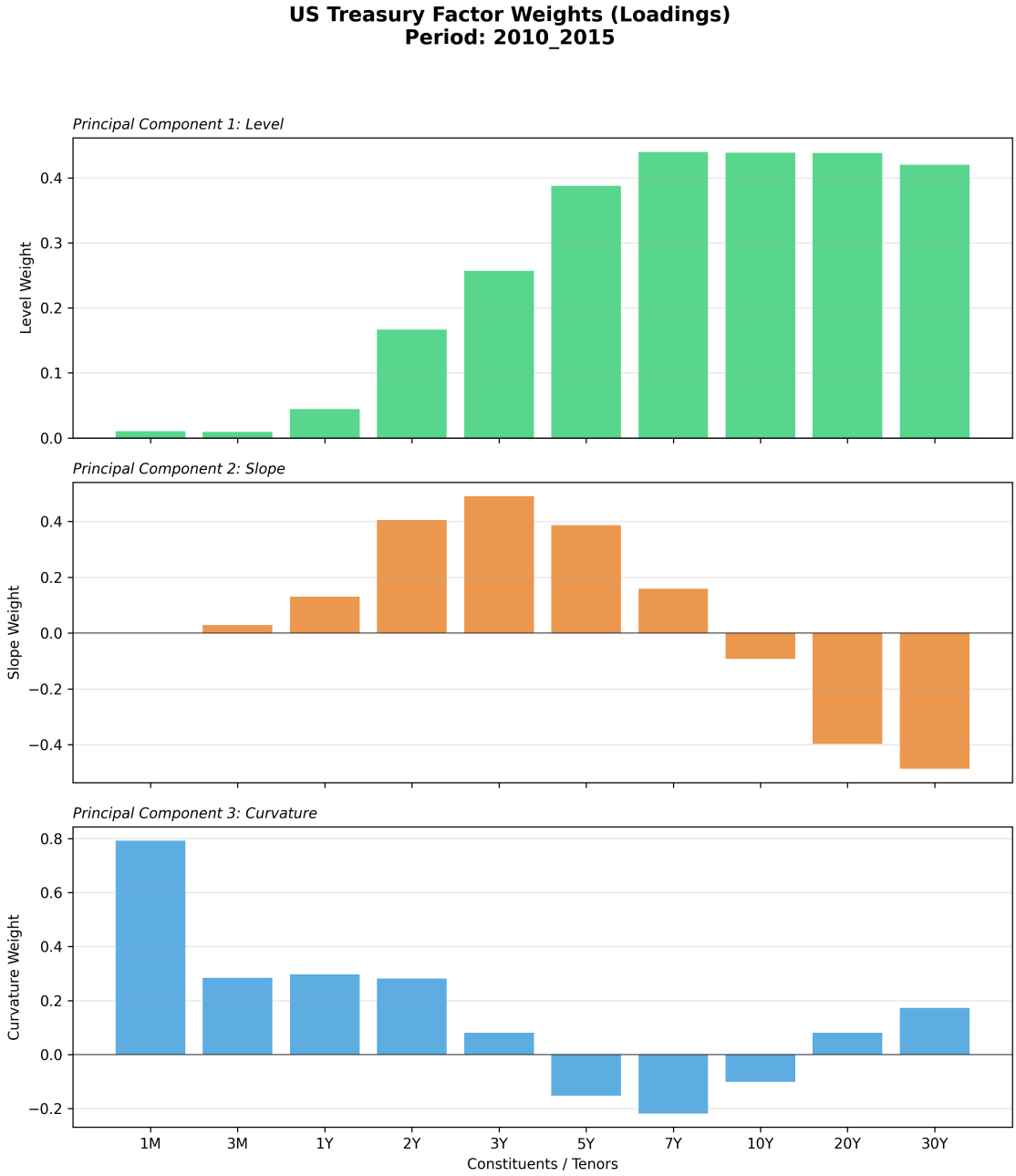

US Treasury Scree Plot: Explained Variance (2015_2020)
Variance Contribution
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
78.3%
8.7%
6.5%
2.9%
1.5%
0.9%
0.4%
0.3%
0.3%
0.2%
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10

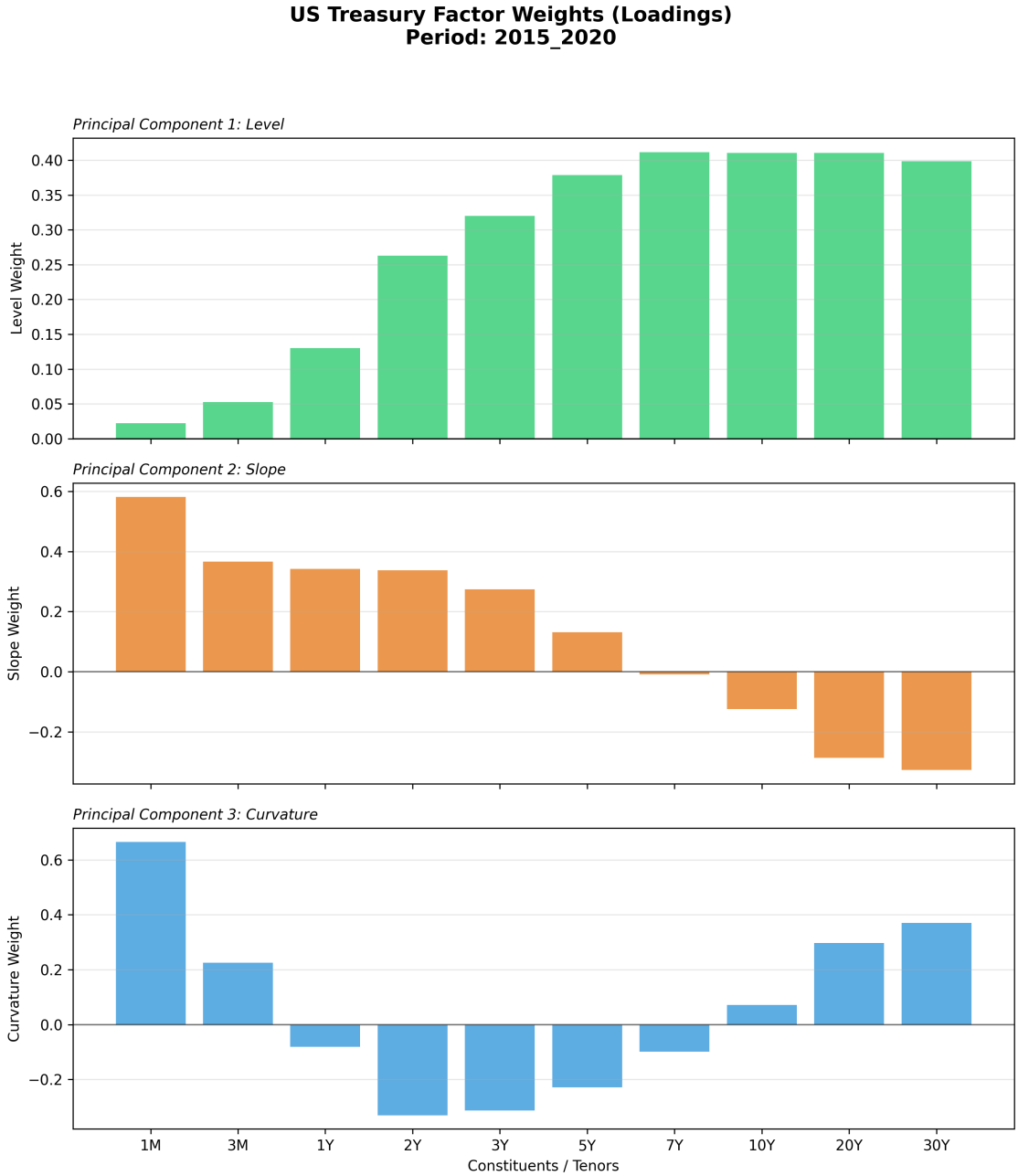

US Treasury Scree Plot: Explained Variance (2020_2025)
Variance Contribution
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
73.2%
10.6%
10.0%
3.4%
1.4%
0.7%
0.3%
0.2%
0.2%
0.1%
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10

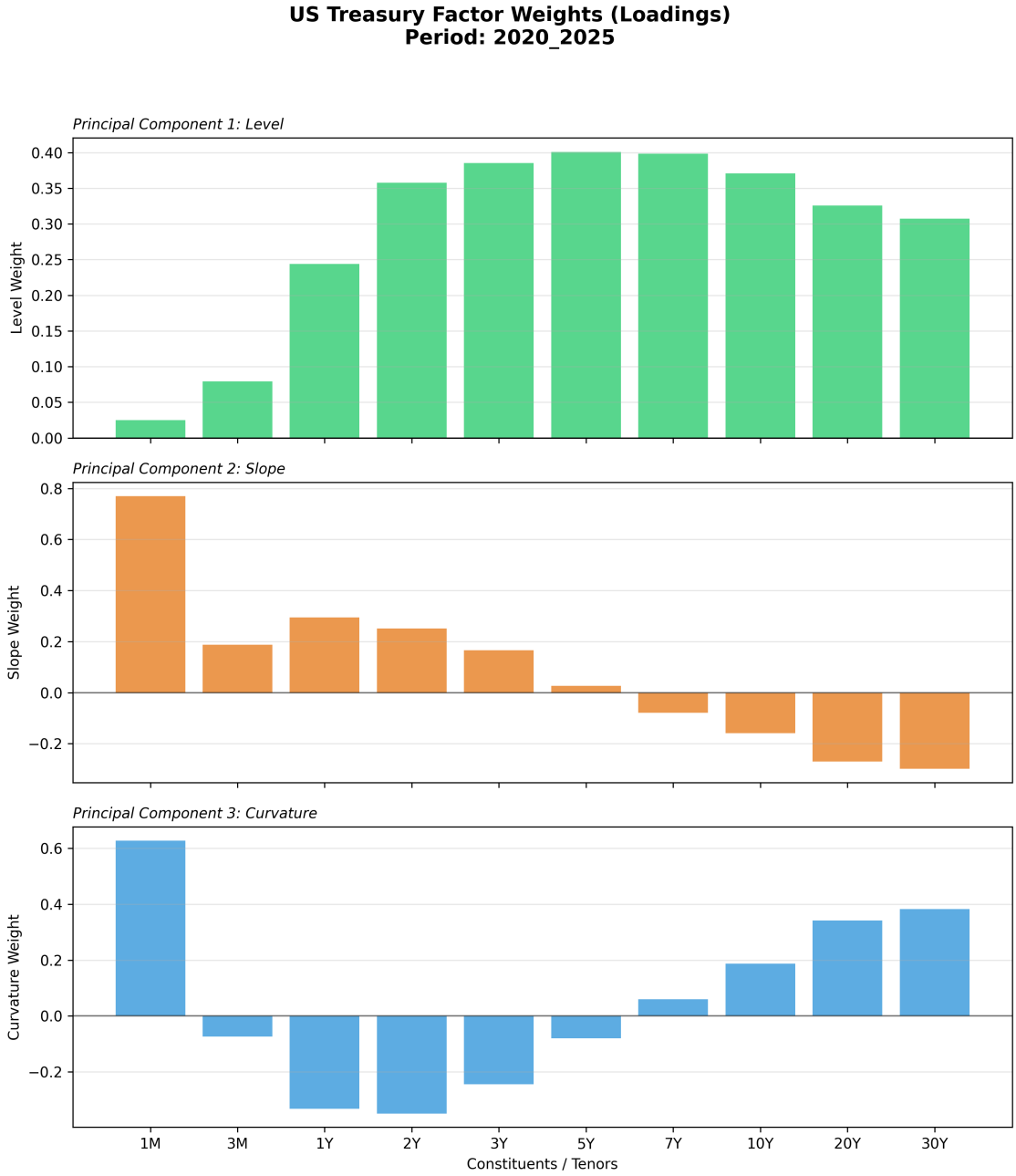
B Interest Rate: Factor Loadings (2000–2025)



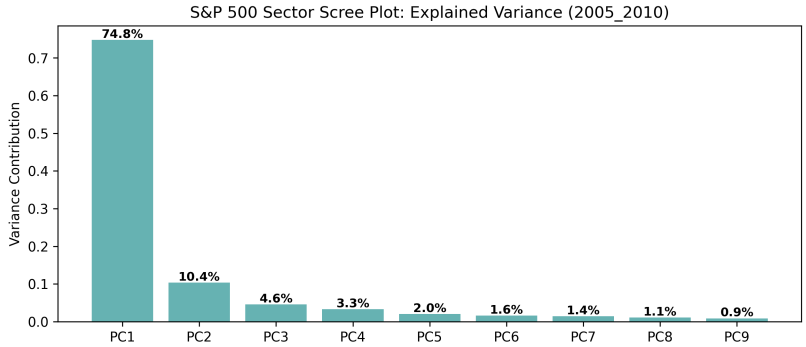

US Treasury Factor Weights (Loadings)
Period: 2005_2010
Principal Component 1: Level
Level Weight
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Principal Component 2: Slope
Slope Weight
0.8
0.6
0.4
0.2
0.0
−0.2
Principal Component 3: Curvature
Curvature Weight
0.4
0.2
0.0
−0.2
−0.4
1M
3M
1Y
2Y
3Y
5Y
7Y
10Y
20Y
30Y
Constituents / Tenors

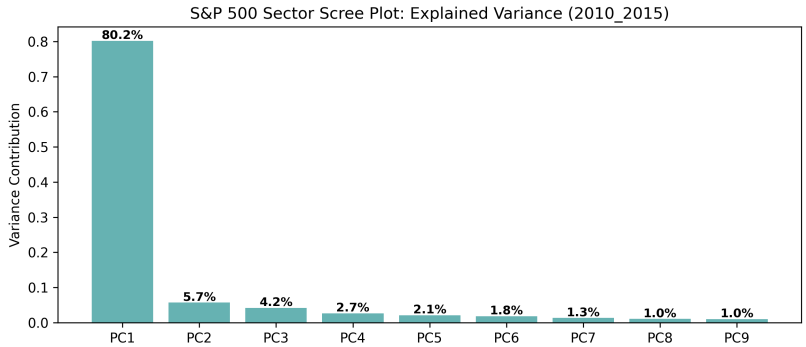

US Treasury Factor Weights (Loadings)
Period: 2010_2015
Principal Component 1: Level
Level Weight
0.0
0.1
0.2
0.3
0.4
Principal Component 2: Slope
Slope Weight
0.4
0.2
0.0
−0.2
−0.4
Principal Component 3: Curvature
Curvature Weight
0.8
0.6
0.4
0.2
0.0
−0.2
1M
3M
1Y
2Y
3Y
5Y
7Y
10Y
20Y
30Y
Constituents / Tenors

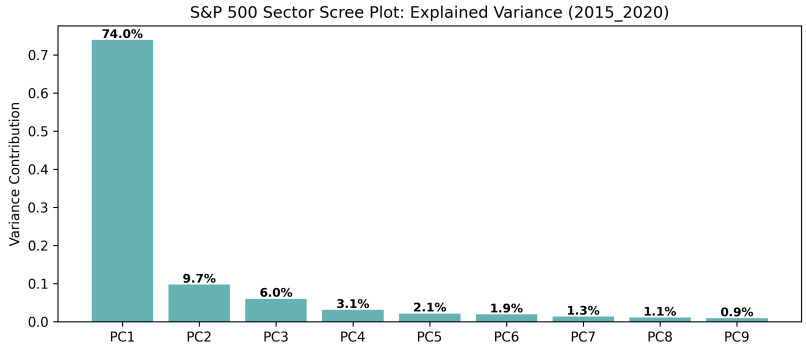

US Treasury Factor Weights (Loadings)
Period: 2015_2020
Principal Component 1: Level
Level Weight
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Principal Component 2: Slope
Slope Weight
−0.2
0.0
0.2
0.4
0.6
Principal Component 3: Curvature
Curvature Weight
−0.2
0.0
0.2
0.4
0.6
1M
3M
1Y
2Y
3Y
5Y
7Y
10Y
20Y
30Y
Constituents / Tenors

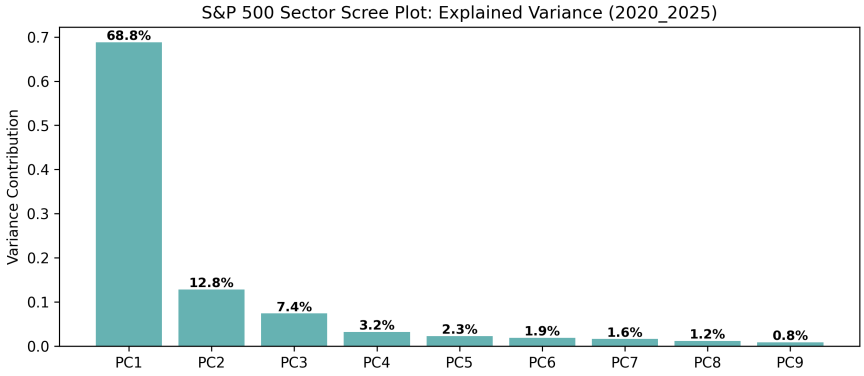

US Treasury Factor Weights (Loadings)
Period: 2020_2025
Principal Component 1: Level
Level Weight
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Principal Component 2: Slope
Slope Weight
0.8
0.6
0.4
0.2
0.0
−0.2
Principal Component 3: Curvature
Curvature Weight
0.6
0.4
0.2
0.0
−0.2
1M
3M
1Y
2Y
3Y
5Y
7Y
10Y
20Y
30Y
Constituents / Tenors

C Equity Sector: Scree Plots (2005–2025)

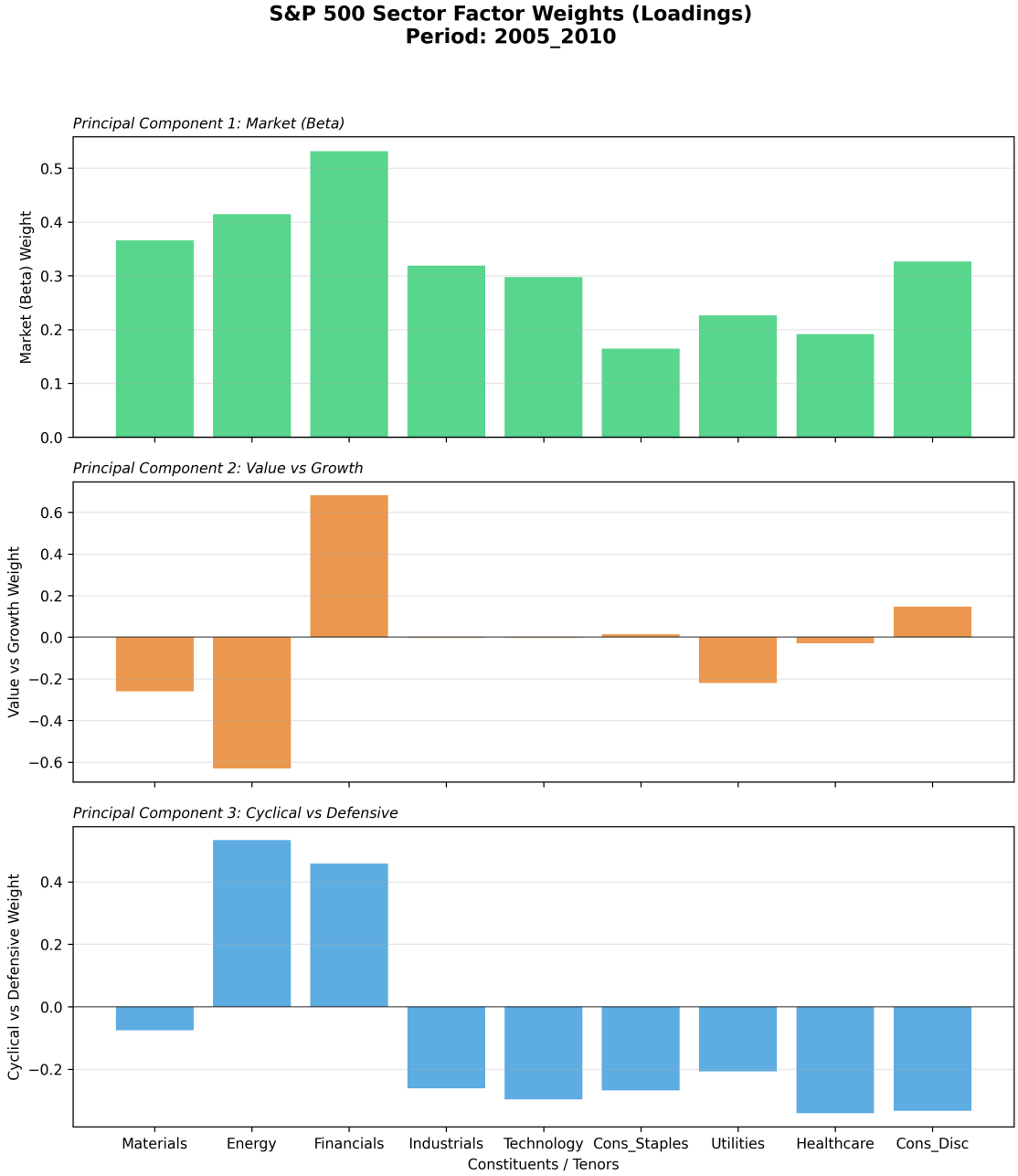




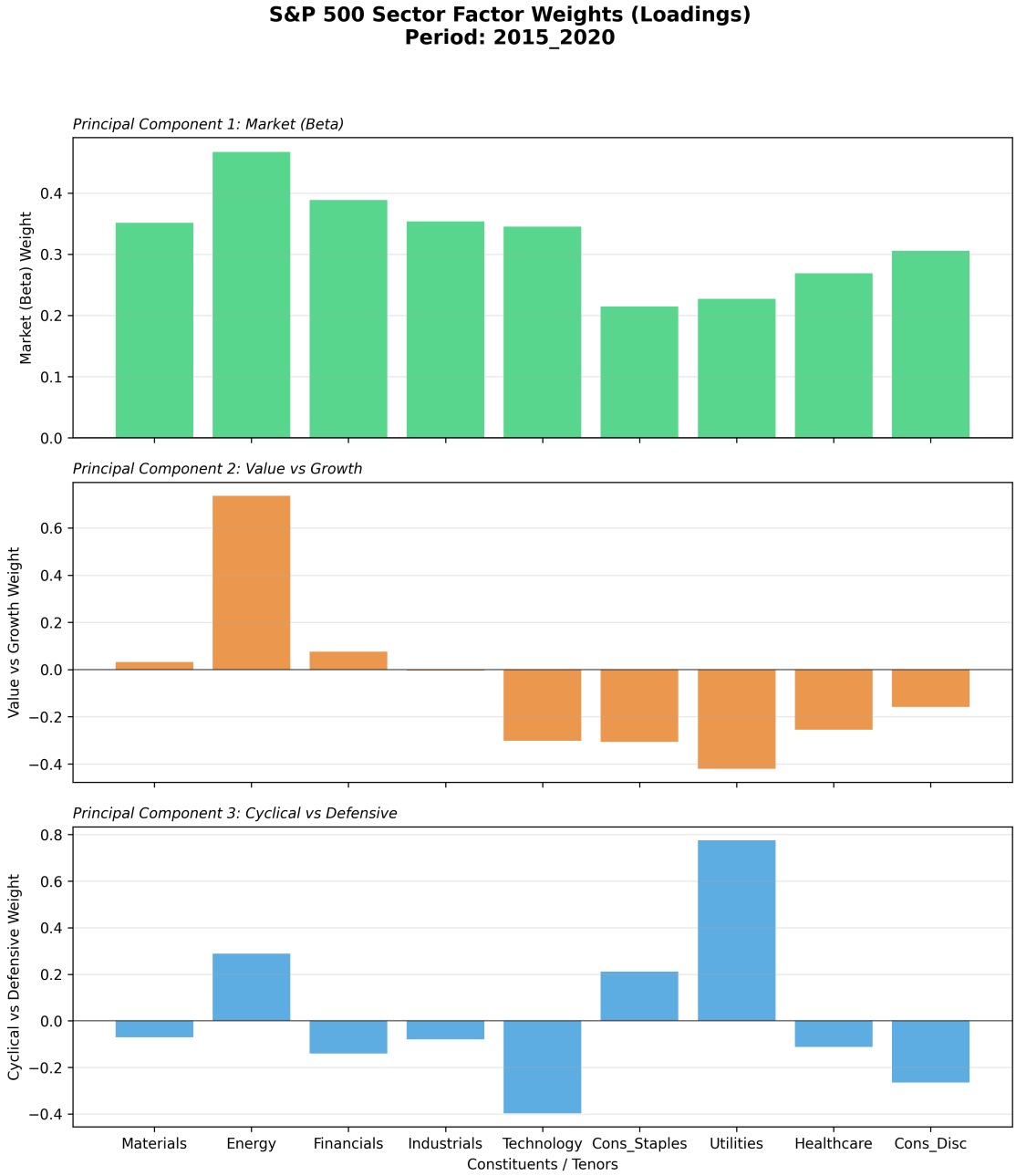



S&P 500 Sector Scree Plot: Explained Variance (2020_2025)
Variance Contribution
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
68.8%
12.8%
7.4%
3.2%
2.3%
1.9%
1.6%
1.2%
0.8%
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9

D Equity Sector: Factor Loadings (2005–2025)



S&P 500 Sector Factor Weights (Loadings)
Period: 2010_2015
Principal Component 1: Market (Beta)
Market (Beta) Weight
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Principal Component 2: Value vs Growth
Value vs Growth Weight
−0.2
0.0
0.2
0.4
0.6
Principal Component 3: Cyclical vs Defensive
Cyclical vs Defensive Weight
−0.2
0.0
0.2
0.4
0.6
0.8
Materials
Energy
Financials
Industrials
Technology
Cons_Staples
Utilities
Healthcare
Cons_Disc
Constituents / Tenors


S&P 500 Sector Factor Weights (Loadings)
Period: 2015_2020
Principal Component 1: Market (Beta)
Market (Beta) Weight
0.0
0.1
0.2
0.3
0.4
Principal Component 2: Value vs Growth
Value vs Growth Weight
−0.4
−0.2
0.0
0.2
0.4
0.6
Principal Component 3: Cyclical vs Defensive
Cyclical vs Defensive Weight
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
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Energy
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Technology
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Cons_Disc
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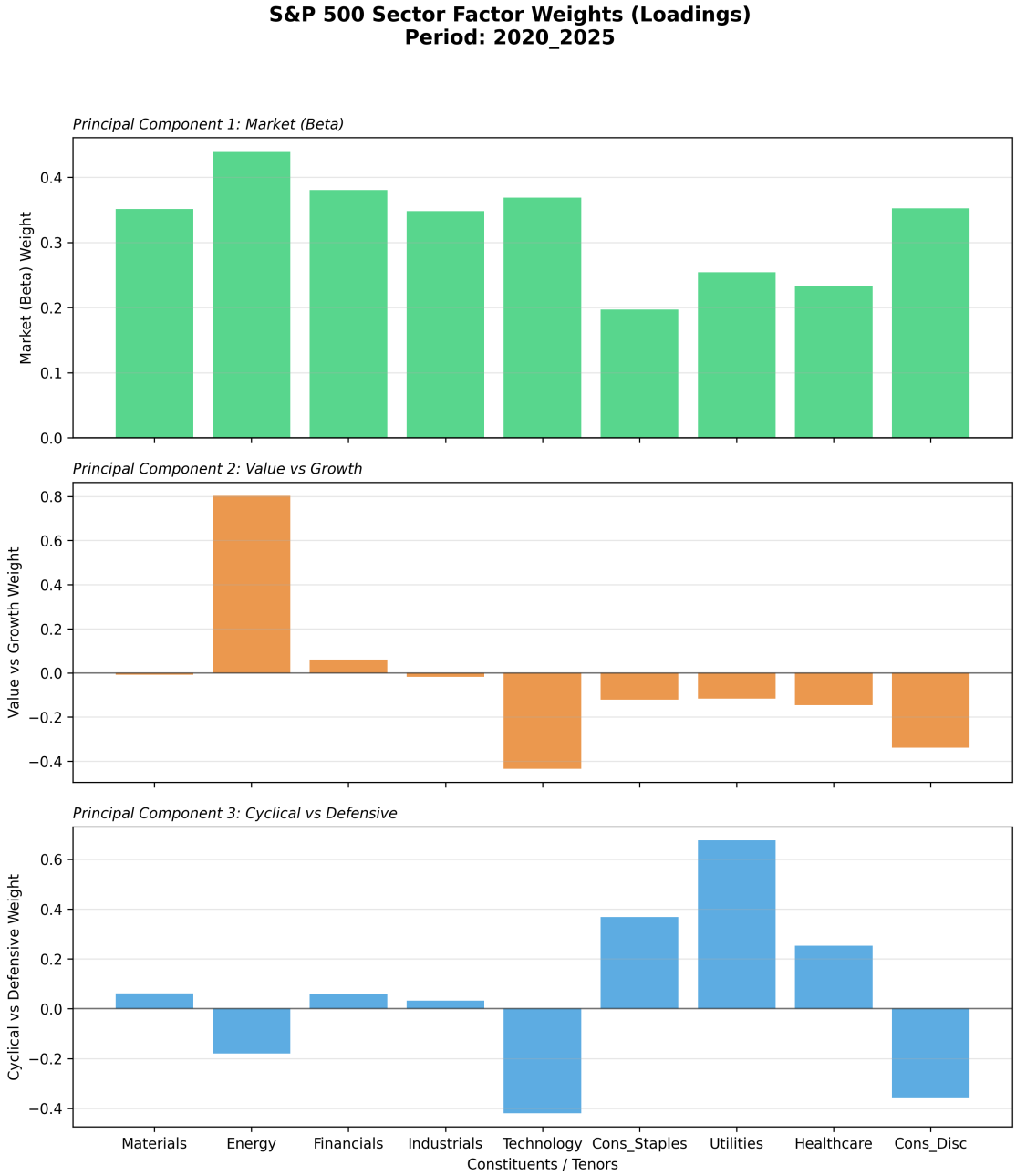
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0.2
0.0
−0.2
−0.4
Principal Component 3: Cyclical vs Defensive
Cyclical vs Defensive Weight
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0.4
0.2
0.0
−0.2
−0.4
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Cons_Disc
Constituents / Tenors